US Economy Time Series Analysis

2026-03-12

Libraries Imported

```
library(ggplot2)
library(dplyr)
library(tidyverse)
library(forecast)
library(strucchange)
```

Importing Data

Trend Analysis

Decomposing to Time Series Components

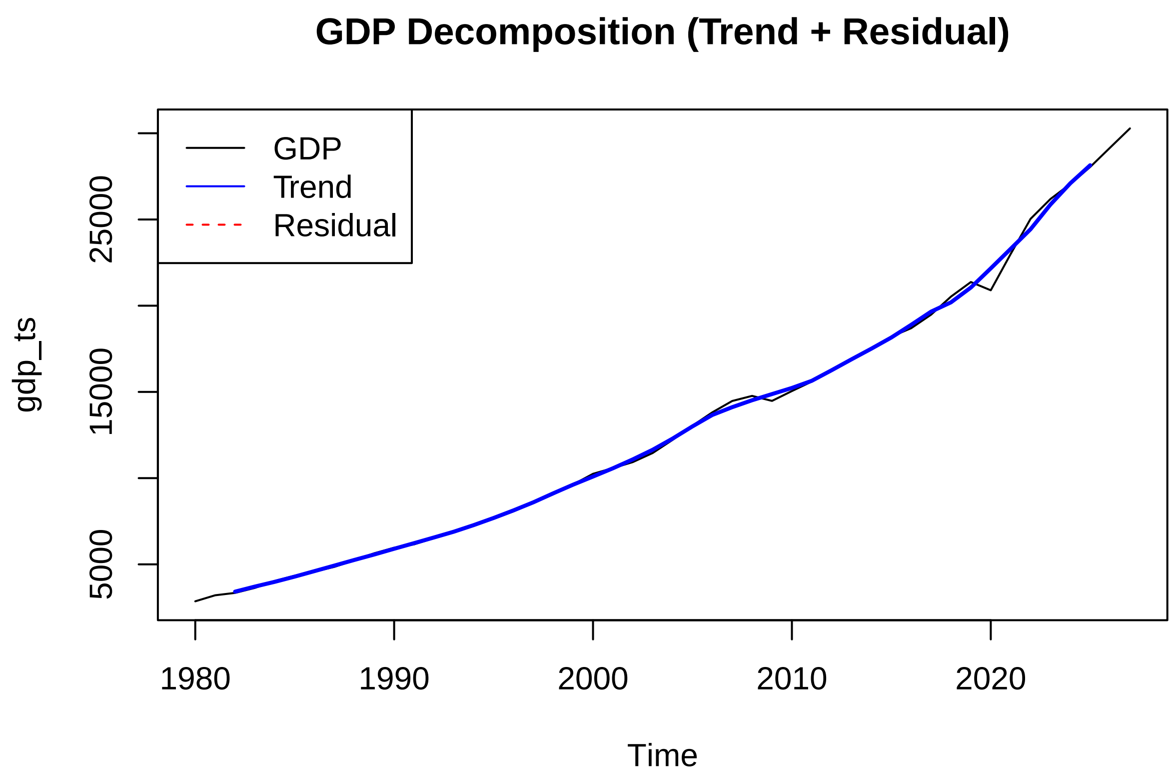
Plotting and Observing Nominal GDP Year By Year

```
# Convert to time series
gdp_ts <- ts(df$GDP_Nominal, start = min(df$Year))

print(gdp_ts)
```

```
## Time Series:
## Start = 1980
## End = 2027
## Frequency = 1
##  [1]  2857.3  3207.0  3343.8  3634.0  4037.7  4339.0  4579.6  4855.3  5236.4
## [10]  5641.6  5963.1  6158.1  6520.3  6858.6  7287.3  7639.8  8073.1  8577.6
## [19]  9062.8  9631.2 10251.0 10581.9 10929.1 11456.5 12217.2 13039.2 13815.6
## [28] 14474.3 14769.9 14478.1 15049.0 15599.7 16254.0 16843.2 17550.7 18206.0
## [37] 18695.1 19479.6 20527.2 21372.6 20893.8 22996.1 25035.2 26185.2 27057.2
## [46] 28045.3 29165.5 30281.5
```

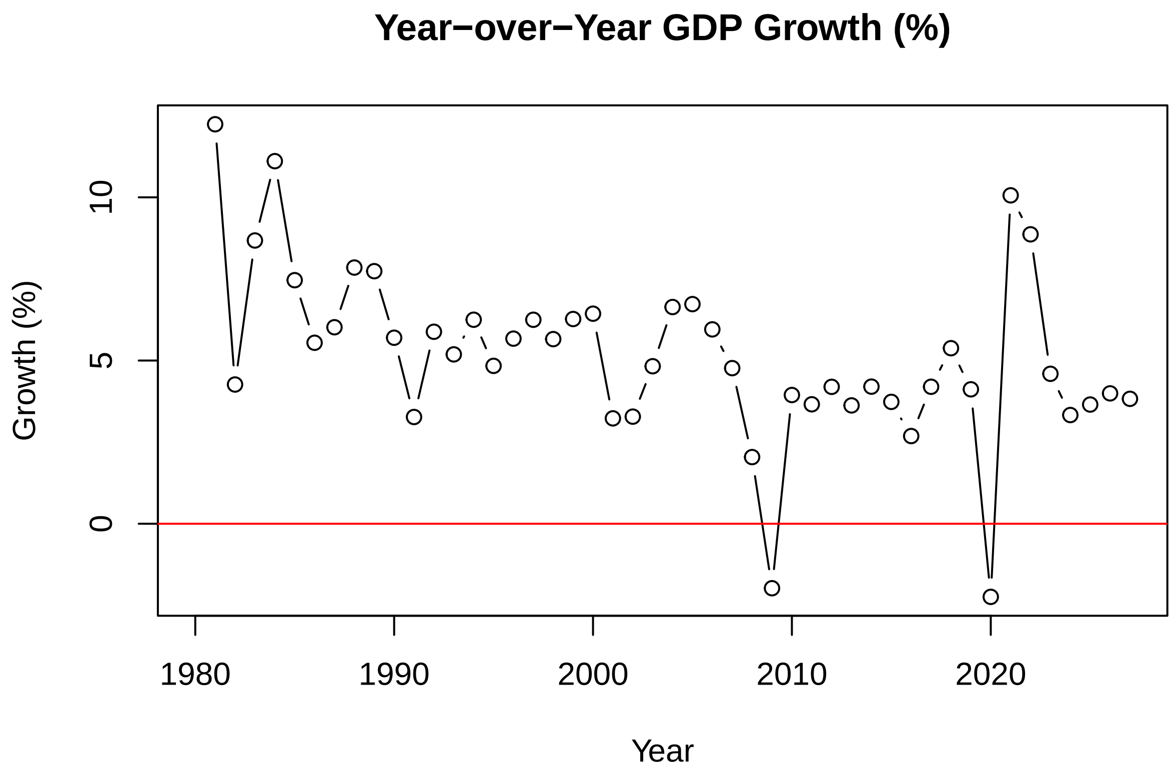
```
trend <- stats::filter(gdp_ts, rep(1/5, 5), sides = 2)

residual <- gdp_ts - trend

# Plotting the Trend and Residuals together, followed by a legend

plot(gdp_ts, main = "GDP Decomposition (Trend + Residual)", col = "black")
```

```
lines(trend, col = "blue", lwd = 2)
lines(residual, col = "red", lty = 2)

legend("topleft",
       legend = c("GDP", "Trend", "Residual"),
       col = c("black", "blue", "red"),
       lty = c(1,1,2))
```



Plotting and Analyzing YoY Growth Rates

```
# Year vector
years <- 1980:2027

# YoY growth
yoy_growth <- c(NA, diff(as.numeric(gdp_ts)) / head(as.numeric(gdp_ts), -1) * 100)

# Plotting Points of Growth Rates Year after Year
plot(years, yoy_growth, type = "b",
     main = "Year-over-Year GDP Growth (%)",
     xlab = "Year", ylab = "Growth (%)")
abline(h = 0, col = "red")
```



In the model above, there is a significant drop in GDP growth at around 2000, which was the year of the Dot-Com Bubble, 2008, which was the year the Great Recession Occurred, and in 2020, which was the year that COVID 19 began. The Great Recession caused a 4.3% decrease in the US GDP. In 2000, there was approximately a 5% GDP drop rate, caused by a significant rise in technology investment. This resulted in an increased unemployment rate, since more jobs at the time were taken over by technology. In 2008, there was a significant increase in unemployment rates, which decreases the amount of business investments. On the other hand, with COVID 19 closing many businesses, it has caused a significant decrease in consumer spending.

Structural Breaks and Regimes

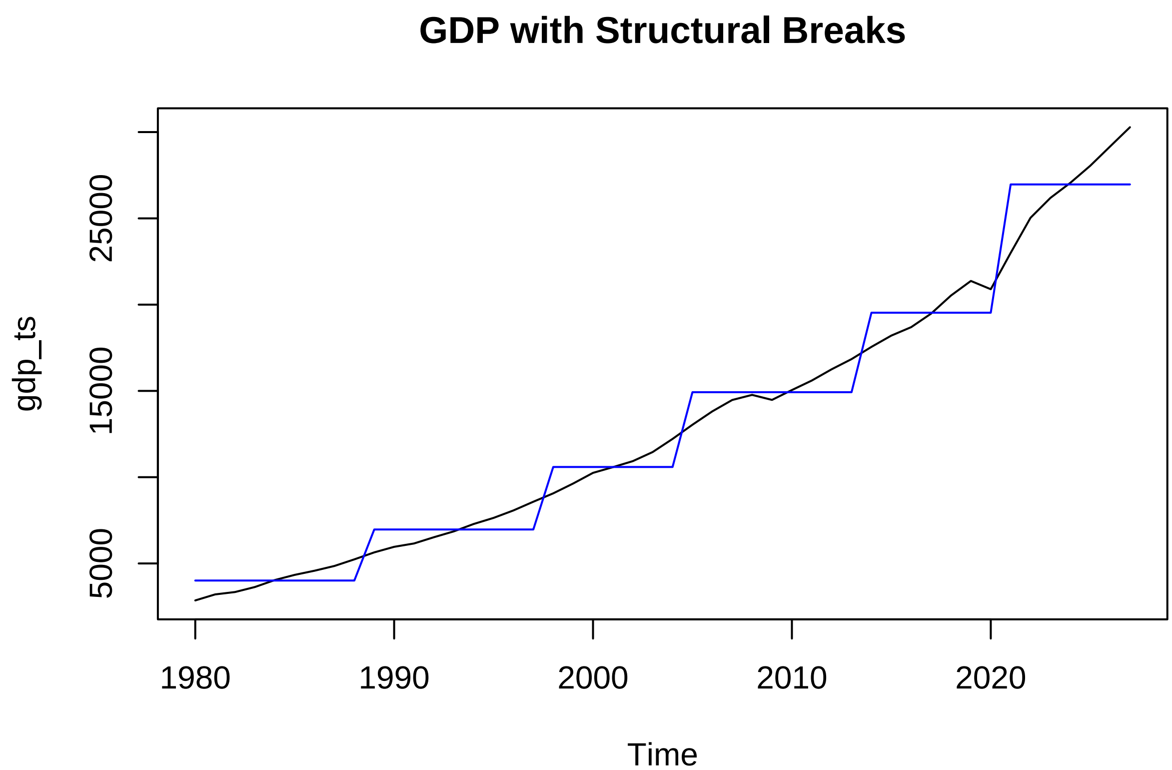
The structural breaks determine if there is any unexpected shifts within the GDP metrics.

```r
# Breakpoint detection
bp_model <- breakpoints(gdp_ts ~ 1)

summary(bp_model)
```

```
##
##   Optimal (m+1)-segment partition:
##
## Call:
## breakpoints.formula(formula = gdp_ts ~ 1)
##
## Breakpoints at observation number:
##
```

```
## m = 1         27
## m = 2         24    41
## m = 3      19    32 41
## m = 4      14 24 34 41
## m = 5   9 18 25 34 41
##
## Corresponding to breakdates:
##
## m = 1              2006
## m = 2              2003      2020
## m = 3         1998      2011 2020
## m = 4         1993 2003 2013 2020
## m = 5    1988 1997 2004 2013 2020
##
## Fit:
##
## m   0         1         2         3         4         5
## RSS 2.872e+09 8.086e+08 3.268e+08 1.790e+08 1.082e+08 8.095e+07
## BIC 1.004e+03 9.504e+02 9.147e+02 8.935e+02 8.771e+02 8.709e+02
```

```r
# Plot breakpoints
plot(gdp_ts, main = "GDP with Structural Breaks")
lines(fitted(bp_model), col = "blue")
```



```r
# Break years
break_years <- time(gdp_ts)[bp_model$breakpoints]
print(break_years)
```

```
## [1] 1988 1997 2004 2013 2020
```

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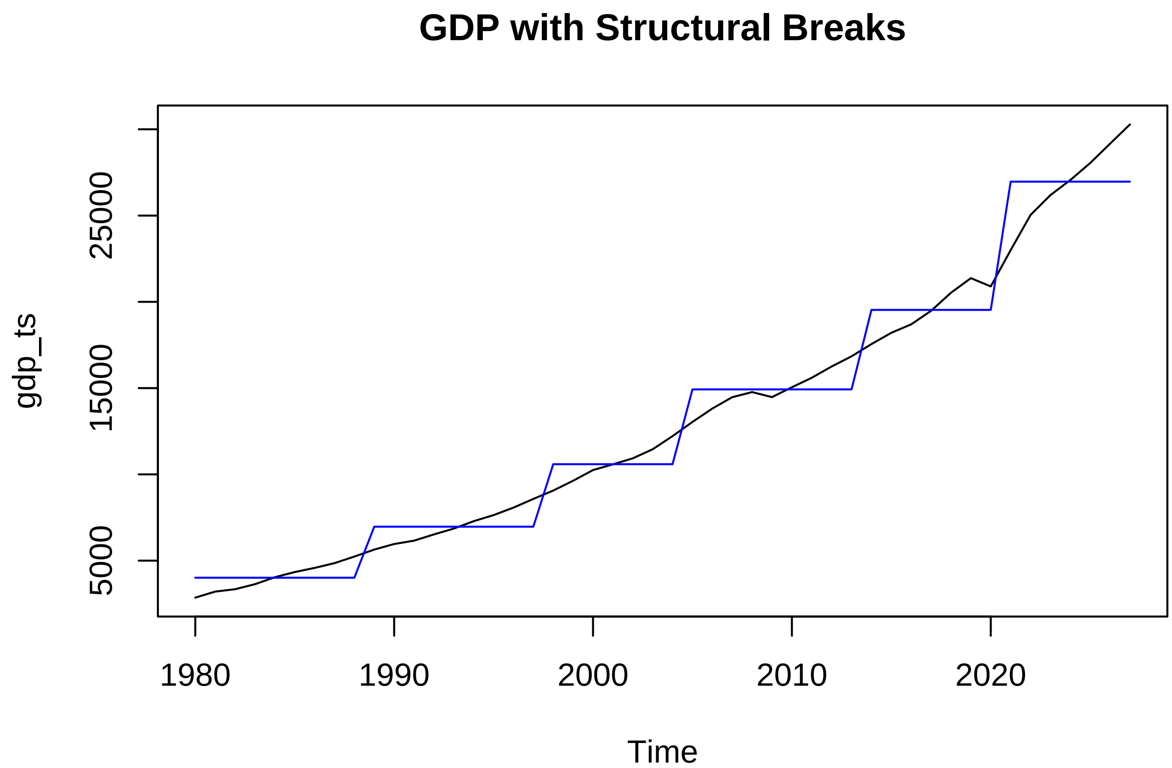


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```
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Here, it can be observed decade by decade, that the Nominal GDP progressively increases, but there is some noticeable, sudden changes around the decades starting at 2000, 2010 and 2020. This is expected, due to the Financial Crisis and COVID-19 events annotated earlier.